

CSE 587B – Project Phase 1 and Phase 2

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Problem Statement:

Given a customer's personal and financial background, estimating the approximate loan amount that can be sanctioned following the statistics in the US Finance sector.

As International students in the US, foreign to the US finance model and primary loan borrowers ourselves, we seek to explore the loan sanction trends in the US subject to the requester. As borrowers, one of the most tedious and lengthy tasks we faced in the loan sanction process was estimating the amount we were eligible for. Thus, we intend to identify these factors in the US loan scenario which takes into account parameters like credit score.

Our goal is to use a comprehensive dataset with the US loan sanction statistics and develop a prediction model that can predict in real time, the loan amount one is eligible for according to their metrics for the deciding factors.

We believe that if such an estimation can be predicted, it will be highly useful for the customer as well as the institution providing the loan.

The problem and why we need to investigate it:

In today's world, which thrives on EMIs and credits, with the boundless opportunities in terms of higher quality education and other amenities like housing, vehicle etc, which are basic yet often a luxury for most people, the ability to afford these often calls for the procurement of a finance loan and this need is ever increasing. But the process in itself is long and complicated, especially for the customer. Traditional processes take days and weeks for the estimation process, affecting the customers who are in immediate need of funds. In addition, the extensive paperwork even before a decision has been made is often very cumbersome and daunting. And as for people new to the system, with the lack of understanding of the finance model in the country, they are faced with a lack of transparency and a lot of uncertainties especially about the basic requirements.

From the banking organization's side, manual processes like these can be a waste of time and

cost, given the redundant nature of the work. The slow decision timeline can also create a negative customer churn for them. It is also likely that due to inaccurate manual assessment, the bank ends up lending loans to high-risk customers, with a high defaulting probability. Making an informed estimation first and quick, and then proceeding with the action comprehensively will be a better decision model, especially in sophisticated finance domains as in the US.

Thus it calls for developing a data driven, statistical model which can provide real time predictions about loan amount eligibility.

Impact of the project and why our contribution is valuable:

Automated eligibility checks reduce the paperwork and long waiting times thus bringing more convenience for the customers. This also improves customer satisfaction and thus creates a positive customer churn for the banking organization as well. With faster information and upfront decisions, customers will have the time and flexibility to explore more options especially when in a time bound scenario like securing an education loan. Also, when presented with such information, it helps the customers plan their finances better. From the banking organization's perspective, an automated, real time, eligible loan amount predictor will help them channel their focus on the potential customers and thus, gives them a competitive edge. It also helps in cutting down cost and time on manual intervention and helps in eliminating the bottlenecks in the lending process. Also, with such data driven decisions, the bank can make quick preliminary analysis and focus better on comprehensive evaluation and thus the probability of lending unmanageable loans is also reduced, which is a risk assessment in favor of the banking organization. The prediction model can also help identify high-value customers, improving targeting strategies. In a country like the US, which has such a vast and diverse population of customers, a faster decision model which is data driven is the way to go, in the field of financial loans.

Data Source: Kaggle- <https://www.kaggle.com/datasets/phileinsophos/predict-loan-amount-data?select=train.csv>

Data Cleaning:

1. Check for missing values

We check for the number of Nan values across all the columns. Two bar graphs were plotted to visualize the number of missing values and percentage of missing values across all the columns in the data.

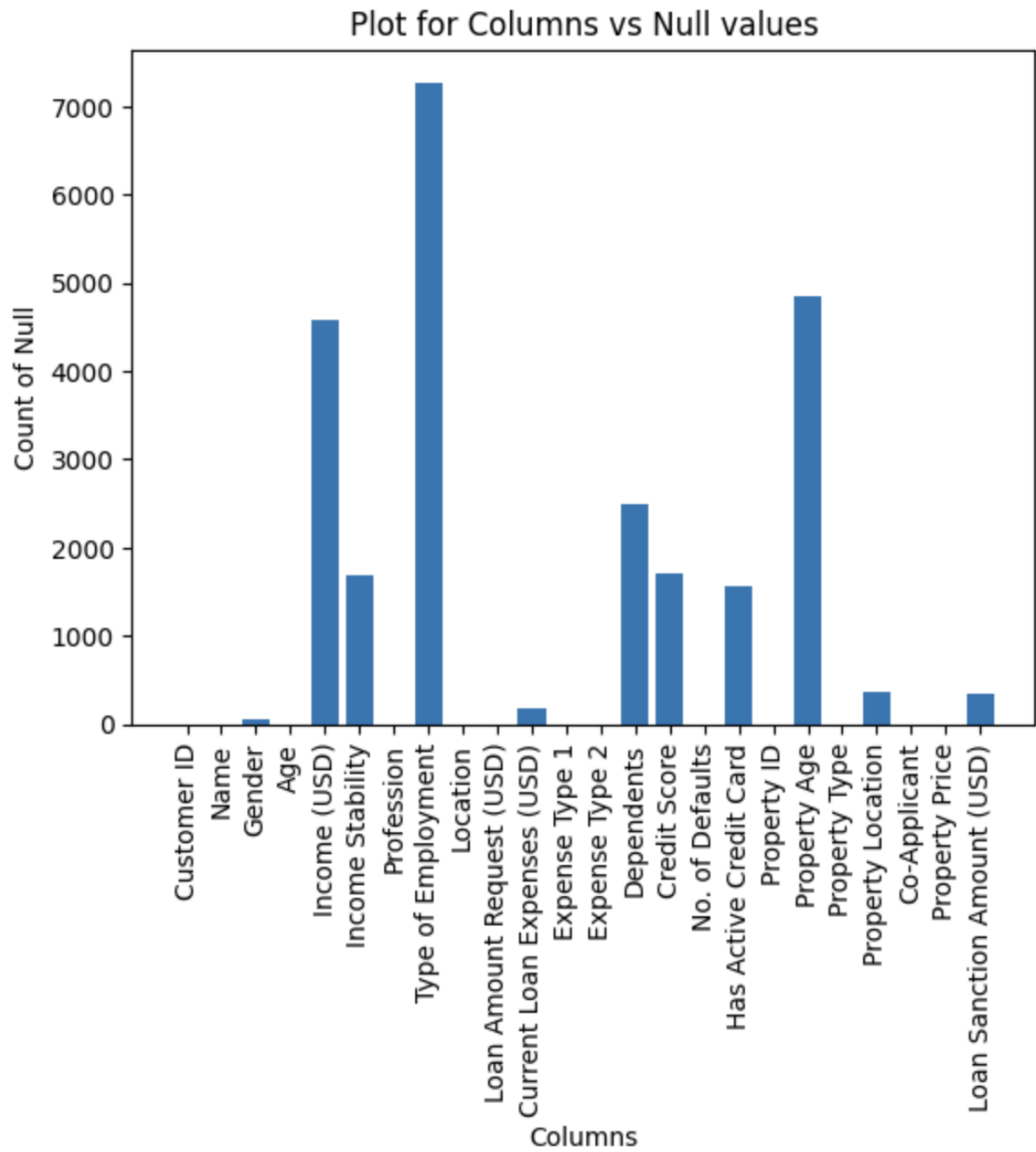


Figure 1: Number of Nan vs Column

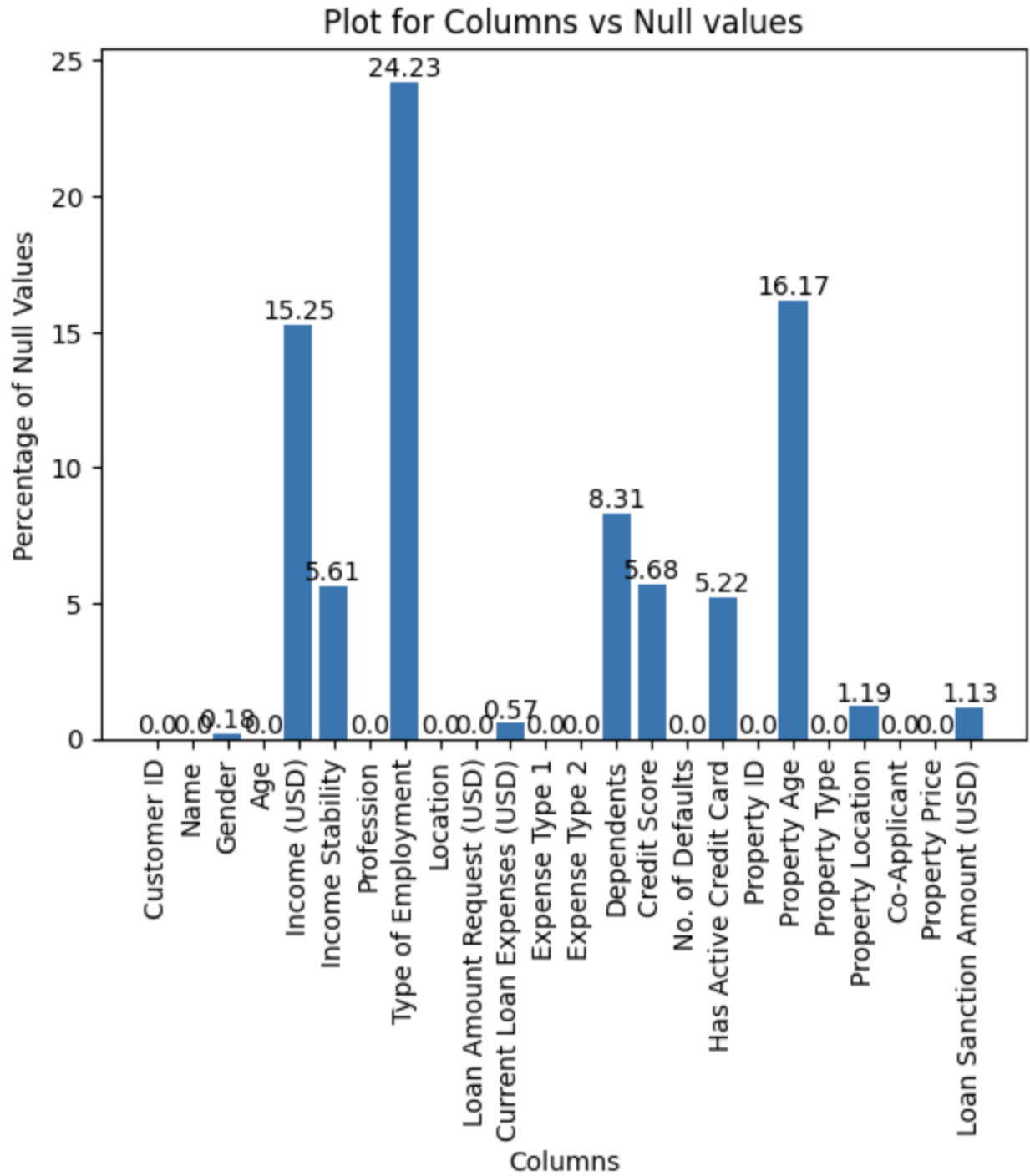


Figure 2: Percentage of Nan vs Column

2.Categorizing Columns

After analyzing the dataset column by column, we categorized the columns into 2 types, numerical and categorical. This separation helps in the further cleaning process as different methods need to be applied for categorical and numerical columns.

- **Numerical columns:**
 - Age
 - Income (USD)
 - Loan Amount Request (USD)
 - Current Loan Expenses (USD)
 - Dependents
 - Credit Score
 - No. of Defaults
 - Property Age
 - Property Price
 - Loan Sanction Amount (USD)
- **Categorical columns:**
 - Gender
 - Income Stability
 - Profession
 - Type of Employment
 - Location
 - Expense Type 1
 - Expense Type 2
 - Property ID
 - Property Type
 - Property Location
 - Co-Applicant
 - Has Active Credit Card

3.Handling missing values in the Target Column

On calculating the percentage of missing or Nan values for the target variable, the value less than 2% whose actual count of records is very less as compared to the entire dataset. Given the small portion, we have decided to drop all the records who have Nan values for target column (Loan Sanction (USD)).

4.Handling missing values in other column

- **Numerical Columns:** Median was used to replace all the Nan values. Median was chosen over mean since the outliers haven't been removed and the outliers may end up skewing the data leading to biased results. Also, the dataset contained many -999 values which are

basically placeholders for missing values. We replaced them with Nan values and continued our processing.

- **Categorical Columns:** Mode was chosen for the replacement of Nan values as they don't have average and using mode helps in preserving the distribution of the data.

5. Handling Outliers

To find the outliers, we find the zscore for each column and replace the records whose absolute zscore value is greater than 3 with their mean. We decided to replace the outliers rather than dropping them as this would lead to a huge reduction in the amount of data leading to the lack of sufficient training data.

Exploratory Data Analysis

1. Univariate Analysis:

This is the simplest form of data analysis, where the data being analyzed consists of just one variable. Since it's a single variable, it doesn't deal with causes or relationships. The main purpose of univariate analysis is to describe the data and find patterns that exist within it. We have performed univariate analysis on multiple features:

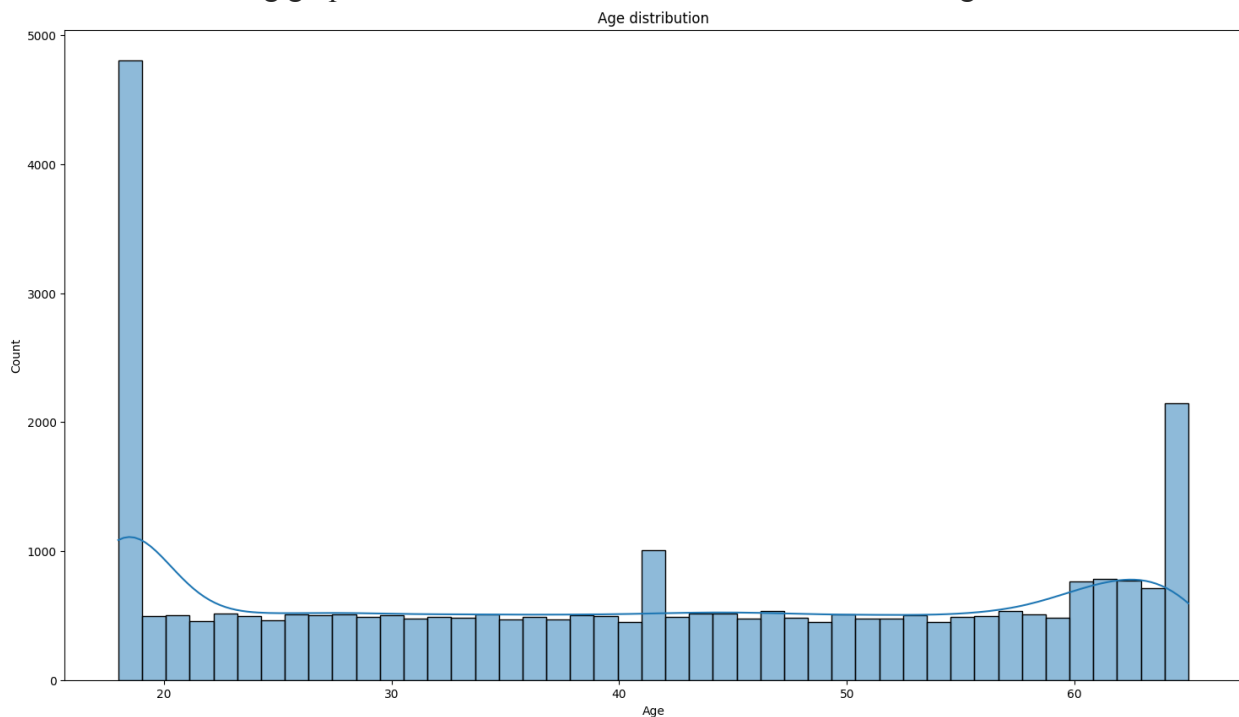
Age:

```
Description:
count      29660.000000
mean        40.099225
std         16.050024
min         18.000000
25%         25.000000
50%         40.000000
75%         55.000000
max         65.000000
Name: Age, dtype: float64
-----
Skewness:
0.046152713177977873
```

- The mean age of a borrower in the given dataset is around 40 years. The age range of the borrowers is from 18-65 years, which indicates a probable minimum and maximum age criterion. The standard deviation is 16 years indicating a good mix of young and old individuals alike. Since the median and mean are both nearly 40 years, the data has minimal skewness and so indicates a symmetric distribution, for the feature Age. With

such symmetry, it makes the feature a good deciding feature in the predictive model we intend to develop.

- The following graph outlines the count distribution of the feature Age.



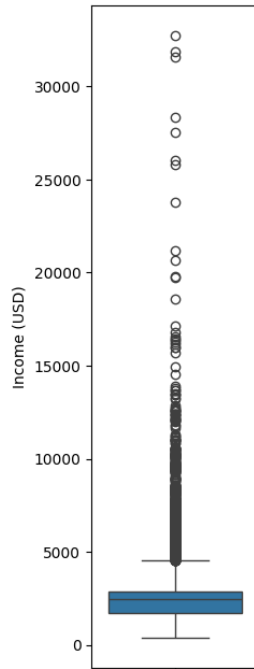
- From the histogram plot we can see that Age is not normally distributed.
- There are many applicants who are 18- and 65-year-old.
- Mean and median of age is 40 years old.
- Skewness is 0.0461 which means the distribution is almost symmetric.

Income:

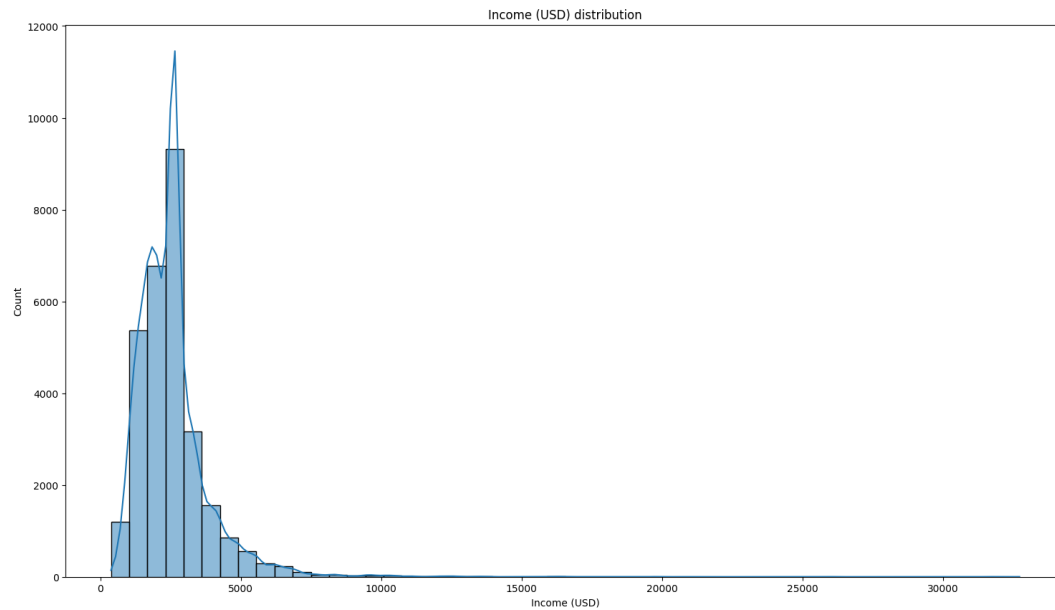
```
Description:
count    29660.000000
mean     2563.807439
std      1360.102977
min       377.700000
25%      1742.322500
50%      2472.265000
75%      2869.327500
max      32726.980000
Name: Income (USD), dtype: float64
-----
Skewness:
4.401922354638985
```

- The mean income is 2563.80 USD. The median or the 50th percentile, is about 2472.27 USD which indicates that mean and median are almost similar which suggests a symmetric distribution of the salary around this range.
- The standard variance for the feature income is 1360.10 USD which suggests a considerably widespread distribution of salaries in the dataset.
- 25% of individuals earn 1,742 USD or less. Half of the population earns below 2,472 USD. 75% of individuals earn below 2,869 USD, meaning most of the population falls into the low to mid-income range. The resultant interquartile range suggests that most of the borrowers fall into the middle-income range.
- The graph below is a boxplot distribution of the feature Income.

Income (USD) distribution (Boxplot)



- The skewness value of 4.40 suggests the positive skew, meaning the distribution has a long tail on the right side of the range.
- The graph below indicates a histogram plot of the feature Income from which we can make these direct inferences:
 - The Income is positively skewed.
 - The median income is \$2472.265.
- The considerable skewness suggests we might need to normalise or standardise the data before we use it for training, to improve model's performance.

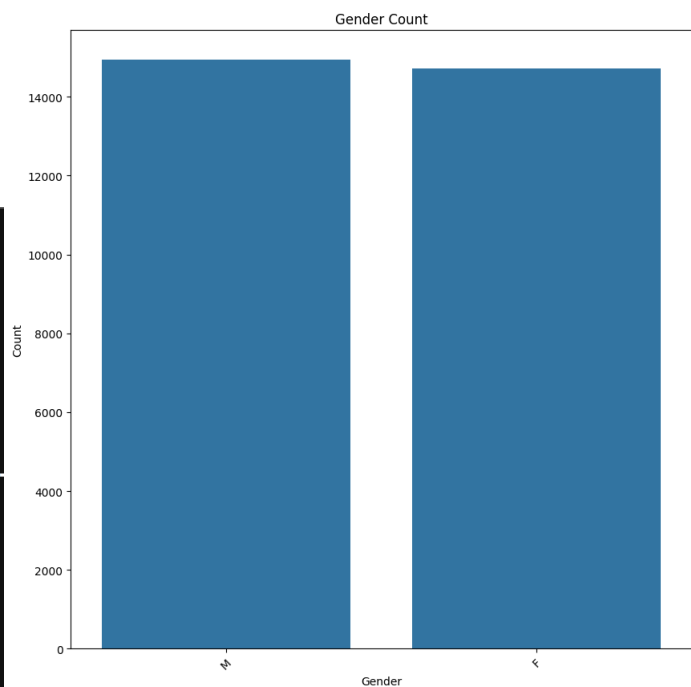


Gender:

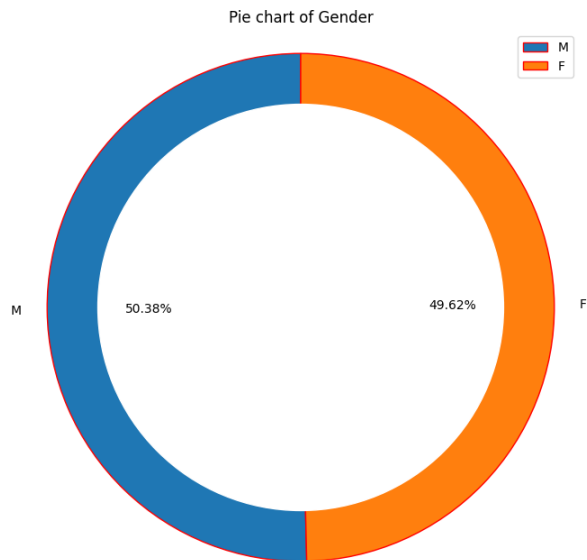
```

Description:
count      29660
unique       2
top         M
freq       14942
Name: Gender, dtype: object
-----
Value count:
      Count  Frequency (%)
Gender
M       14942      50.377613
F       14718      49.622387

```



- The distribution of the feature gender suggests that most of the borrowers are Males by a marginal difference. The gender female is almost equal in numbers as male borrowers.



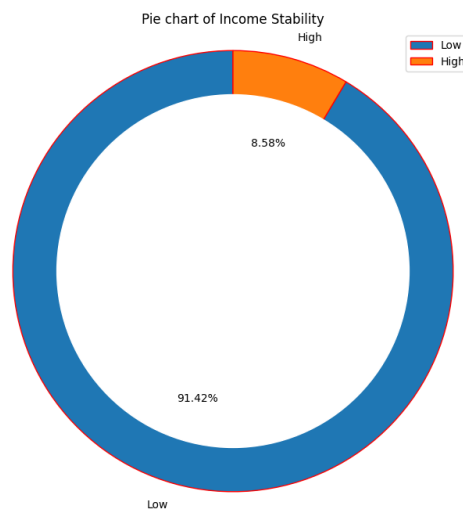
Income Stability:

```

Description:
count      29660
unique       2
top         Low
freq       27116
Name: Income Stability, dtype: object
-----
Value count:

```

	Count	Frequency (%)
Income Stability		
Low	27116	91.422792
High	2544	8.577208



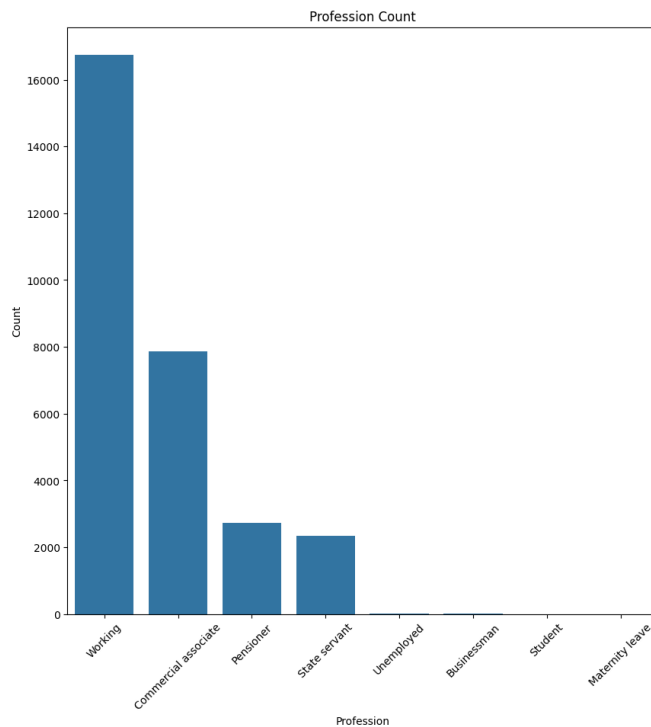
- The distribution of the feature income stability suggests that most of the borrowers in the given dataset falls in the low-income stability category.
- Less than 10 percent only falls in the high-income stability category.
- This suggests that the banks may rely on additional factors more than income stability to assess eligibility.

Profession:

```
Description:
count      29660
unique       8
top      Working
freq      16739
Name: Profession, dtype: object
-----
Value count:

```

	Count	Frequency (%)
Profession		
Working	16739	56.436278
Commercial associate	7863	26.510452
Pensioner	2718	9.163857
State servant	2334	7.869184
Unemployed	2	0.006743
Businessman	2	0.006743
Student	1	0.003372
Maternity leave	1	0.003372



Interpretation:

- Most applicants are currently working, the second most represented category is commercial associate.
- We also have several applicants that have a pension or are state servants.
- The most unrepresented categories are the unemployed, businessman, students and women who are on maternity leaves.

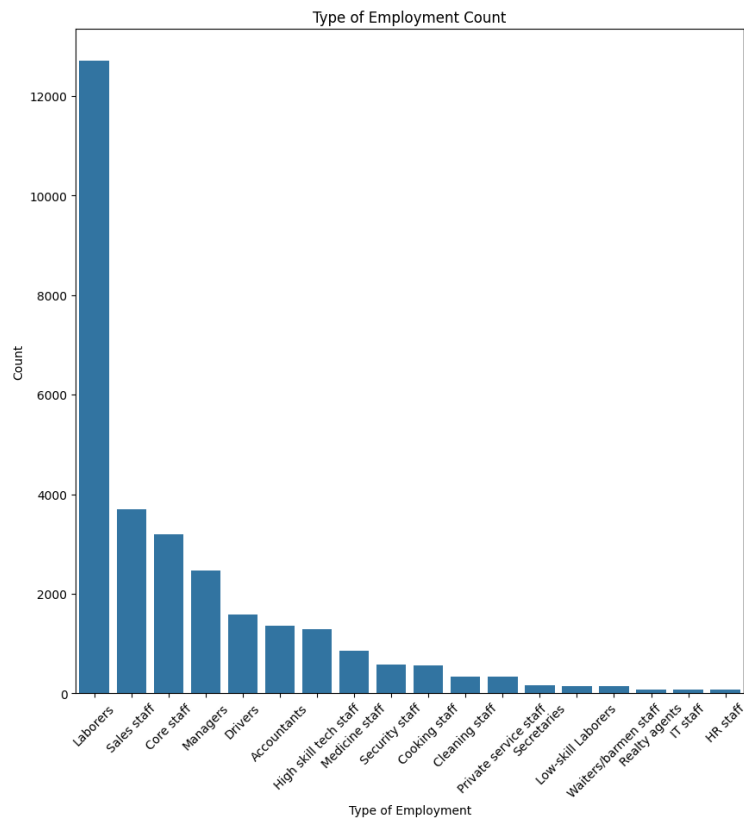
Type of Employment:

```

Description:
count      29660
unique       18
top      Laborers
freq      12704
Name: Type of Employment, dtype: object
-----
Value count:

```

Type of Employment	Count	Frequency (%)
Laborers	12704	42.832097
Sales staff	3698	12.467970
Core staff	3189	10.751854
Managers	2464	8.307485
Drivers	1588	5.354012
Accountants	1363	4.595415
High skill tech staff	1297	4.372893
Medicine staff	854	2.879299
Security staff	574	1.935266
Cooking staff	561	1.891436
Cleaning staff	338	1.139582
Private service staff	337	1.136210
Secretaries	158	0.532704
Low-skill Laborers	154	0.519218
Waiters/barmen staff	148	0.498989
Realty agents	85	0.286581
IT staff	77	0.259609
HR staff	71	0.239380



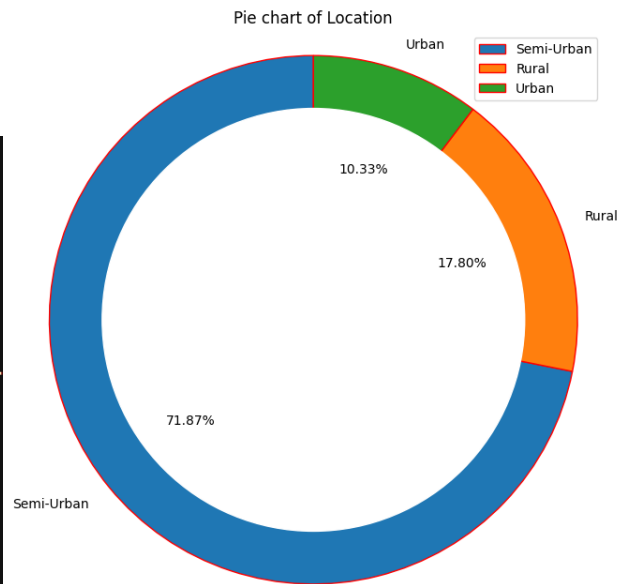
- The data suggests that most of the loan applicants have manual or semi-skilled jobs. The distribution of high skill jobs is sparse in the given data of applicants. This suggests that the applicants might come from a low- to middle-income demographic, with a strong representation of working-class individuals. The imbalance in job types (with laborers dominating) could affect model performance. This feature will require careful placement of weights to provide a model with efficient prediction.

Location:

```

Description:
count      29660
unique       3
top      Semi-Urban
freq      21317
Name: Location, dtype: object
-----
Value count:
      Count  Frequency (%)
Location
Semi-Urban  21317      71.871207
Rural       5280      17.801753
Urban       3063      10.327040

```



- The distribution of the feature location suggests most of the loan requestors come from semi-urban areas. And the least from Urban areas.

Loan Amount Requested:

Description:

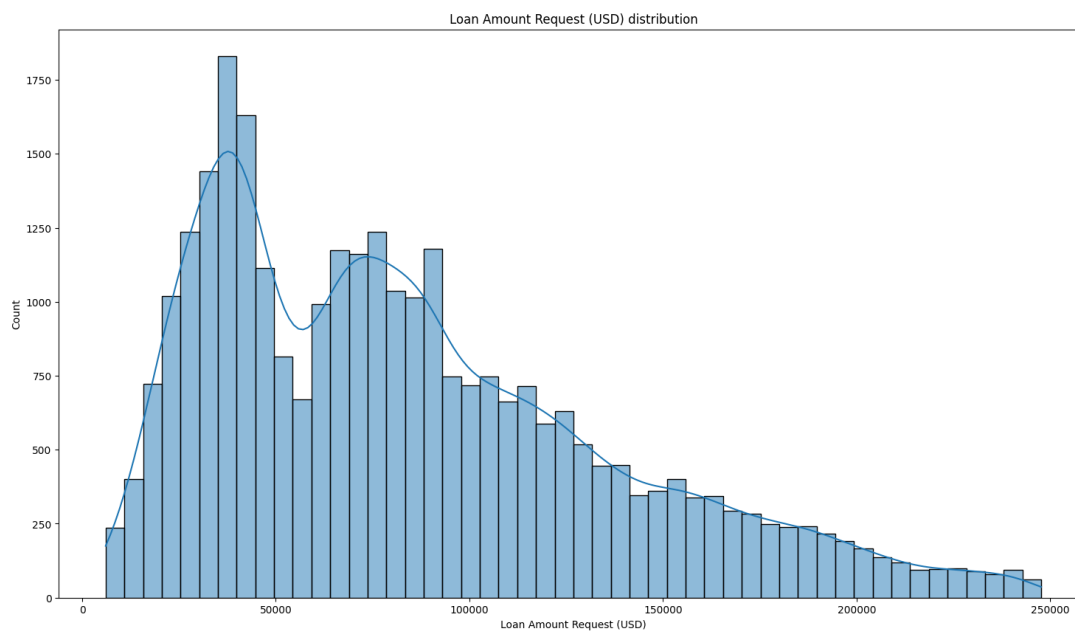
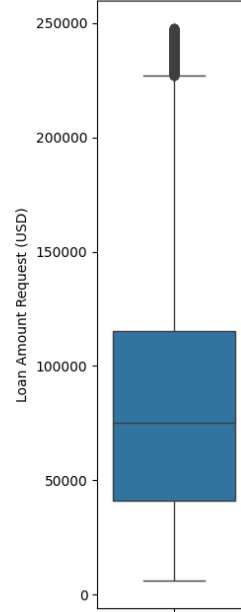
```
count    29660.000000
mean      84803.467296
std       51977.735492
min        6048.240000
25%       41200.112500
50%       75114.085000
75%      115454.802500
max      247671.200000
```

```
Name: Loan Amount Request (USD), dtype: float64
```

Skewness:

```
0.8394578123410887
```

Loan Amount Request (USD) distribution (Boxplot)



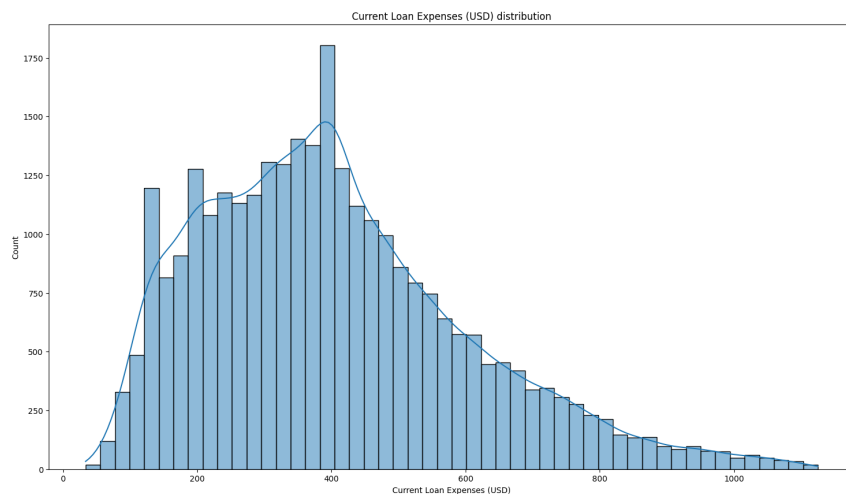
- On average, individuals request around 84,800 USD in loans. The median is slightly lower than the mean, indicating that a few large loan requests are pulling the mean upwards.
- The standard deviation suggests that there is significant variability in the loan requests, with amounts spread widely across different values. This also suggests the dataset covers both small and large loan requests.
- 25% of loan requests are 41,200 USD or less. Half of the loan requests are 75,114 USD or less. 75% of loan requests are 115,454 USD or less, meaning the majority of loan requests are in the small to mid-sized range.
- Interquartile Range (IQR): 74,254.69 USD (115,454.80 - 41,200.11) indicates that typical spread of loan requests is in the middle 50% of the population.
- Although the skewness is not extreme (0.839), this feature will require normalization techniques to improve model performance.

Current Loan Expenses:

```

Description:
count      29660.000000
mean        402.773108
std         198.697025
min         33.760000
25%         252.270000
50%         379.310000
75%         515.100000
max         1125.100000
Name: Current Loan Expenses (USD), dtype: float64
-----
Skewness:
0.773214080223501

```



- On average, individuals are paying 402.77 USD in existing loan-related expenses. The median is slightly lower than the mean, which suggests mild positive skew—a few higher expenses may be driving the mean upward. The variation in loan expenses is moderate, indicating some individuals have significantly higher loan obligations. Interquartile Range (IQR): 262.83 USD (515.10 - 252.27) suggests most individuals have loan payments that fall between 252.27 USD and 515.10 USD.

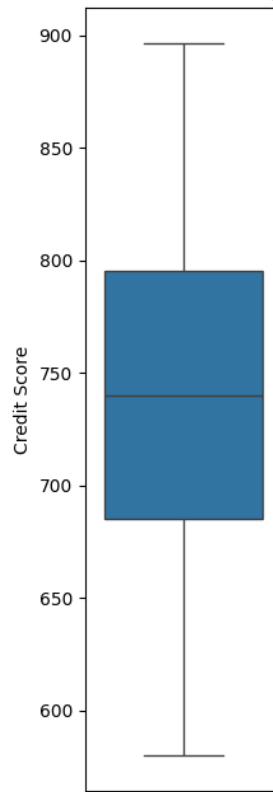
Credit Score:

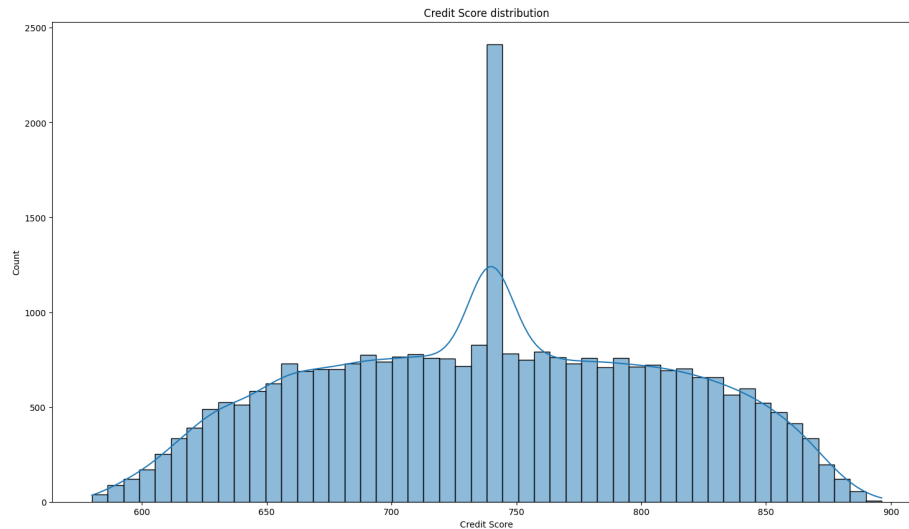
```

Description:
count      29660.000000
mean        739.853475
std         70.091375
min         580.000000
25%         685.317500
50%         739.853475
75%         795.120000
max         896.260000
Name: Credit Score, dtype: float64
-----
Skewness:
-0.0208527715797445

```

Credit Score distribution (Boxplot)





- The average credit score is around 740, which is considered good by most credit scoring standards (e.g., FICO). Since the mean and median are identical, the distribution is symmetrical with no significant outliers. The standard deviation of 70.09, suggests a moderate spread of credit scores, with most scores clustered around the mean of 740. The highest credit score is nearly 900, indicating some borrowers have exceptional creditworthiness. With an average credit score around 740 and a symmetrical distribution, most individuals in this dataset have good to very good credit which suggests that a good credit score is probably one of the strongest determinants of eligible loan amount.

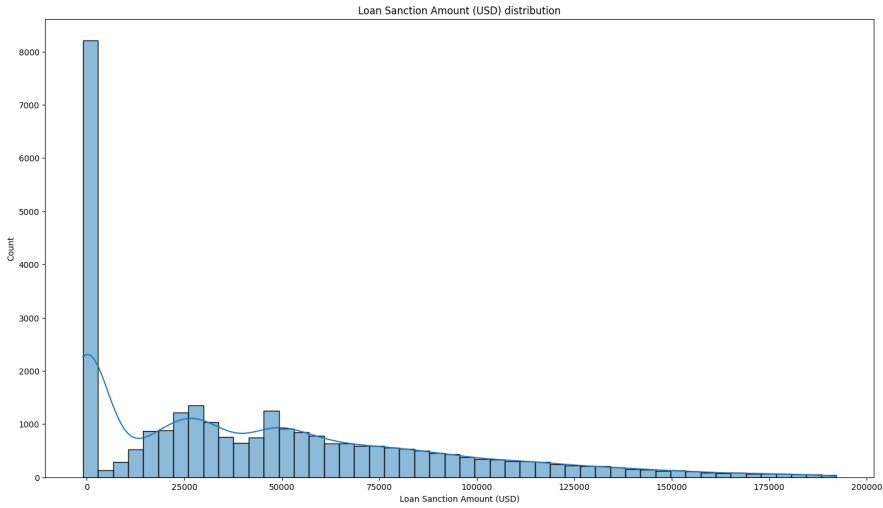
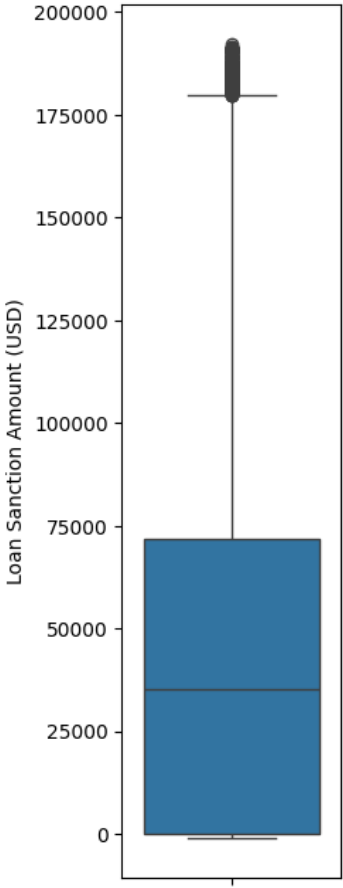
Loan Sanction Amount:

```

Description:
count      29322.000000
mean       46065.779403
std        43990.166350
min         0.000000
25%        0.000000
50%        36064.225000
75%        72535.995000
max        192824.800000
Name: Loan Sanction Amount (USD), dtype: float64
-----
Skewness:
0.9091874671019101

```

Loan Sanction Amount (USD) distribution (Boxplot)



- On average, the sanctioned loan amount is about 45,503 USD. The median is lower than the mean, suggesting the distribution is positively skewed (a few high values increase the mean).
- The large standard deviation indicates high variability in the loan amounts sanctioned.

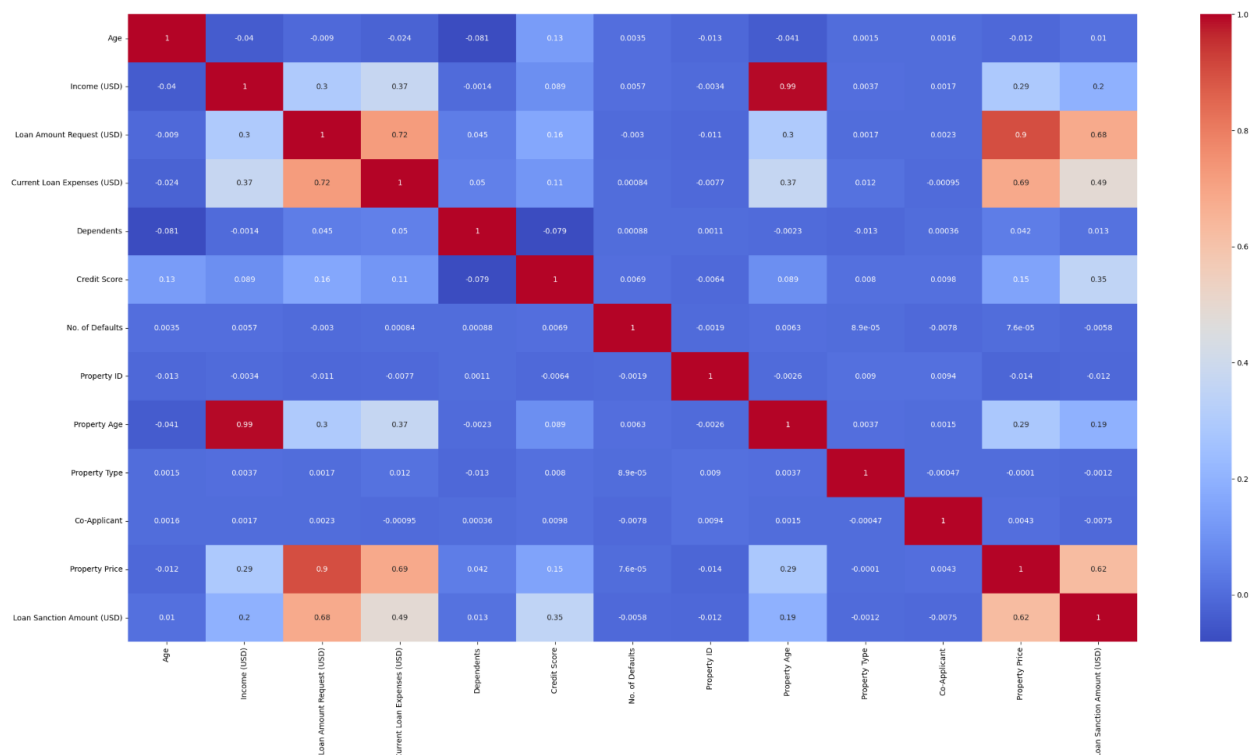
2. Multivariate Analysis:

Multivariate data arises from more than one variable. It uses graphics to display relationships between two or more sets of data. It focuses on understanding relationships and patterns across multiple variables simultaneously. The goal of multivariate analysis is to understand how variables interact with each other and how their combined influence impacts an outcome or a set of outcomes.

Correlation Matrix – Heatmap

A correlation matrix displays the pairwise correlation coefficients between variables. Each correlation value ranges between -1 to 1 , with -1 equivalent to inverse proportionality and 1 direct proportionality.

Using a correlation matrix, we can conclude which features positively impact the dependent variable Y .



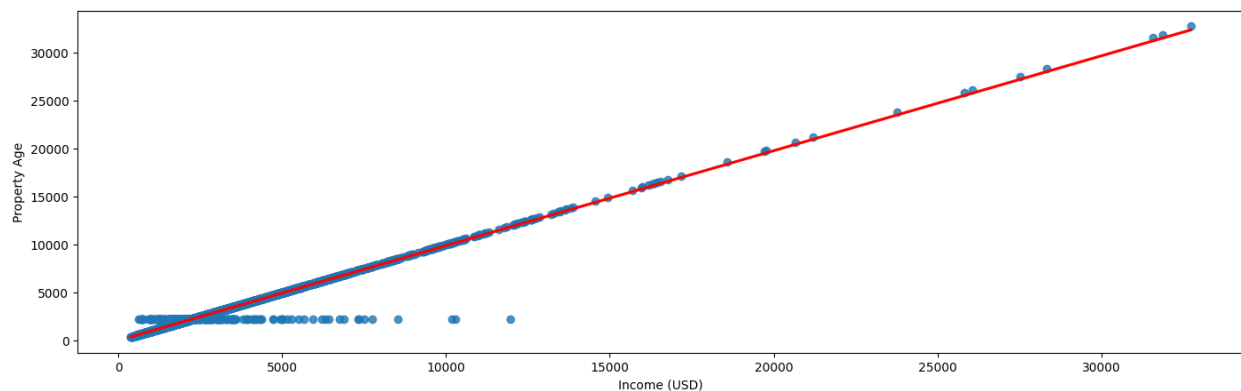
From the above matrix, we can make 5 main conclusions based on large correlation coefficients -

- Income is highly correlated with Property Age ~ 99%
- Loan Amount Requested is highly correlated with Property Price ~ 90%
- Loan Amount Requested is decently correlated with Current Loan Expenses ~ 72%
- Current Loan Expenses is decently correlated with Property Price ~ 69%
- Loan Amount Requested (USD) is directly correlated with the Loan Sanction Amount (USD)

Scatter Plots -

We next individual check the pairwise correlations for the above interpretations using scatter plots -

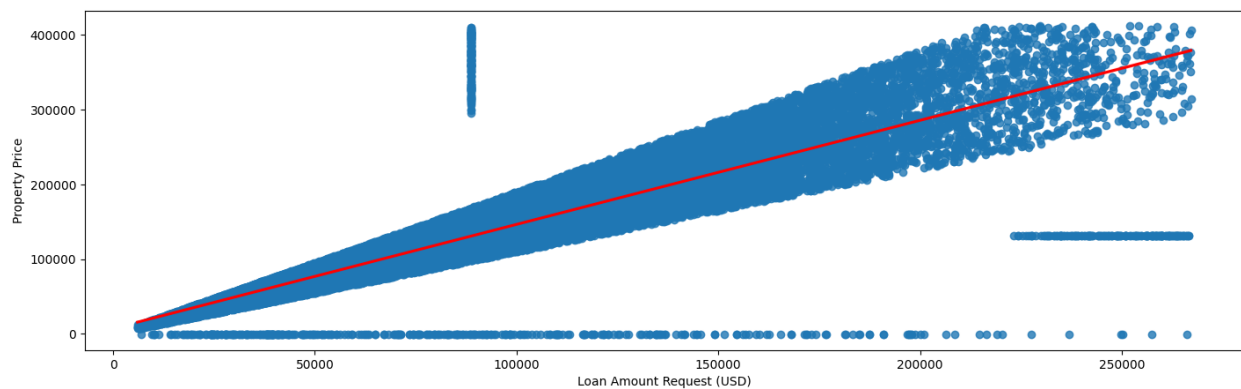
Income vs Property Age



From the above graph we can inference that Income and Property Age are almost closely equivalent. As the property ages, the income of the buyer seems to be higher. A very good correlation between the two.

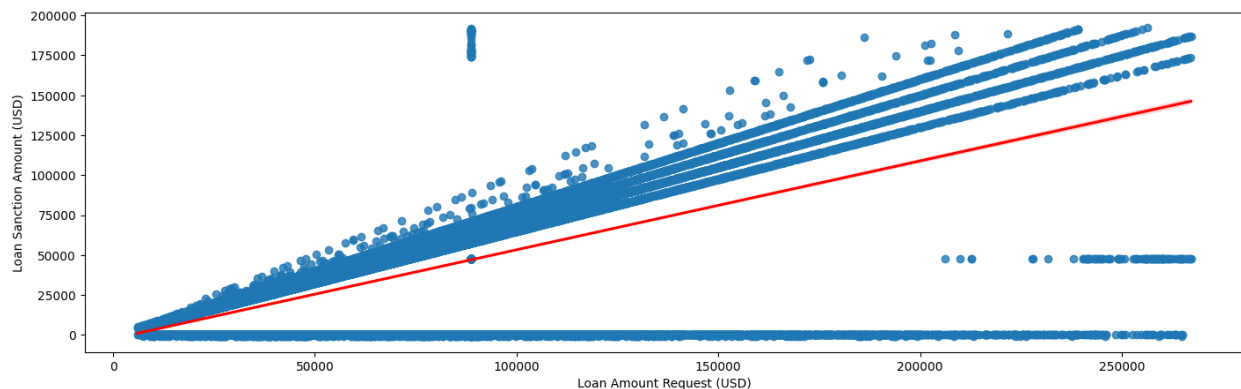
We can infer that high income families like to stay and buy older property's.

Loan Amount Requested vs Property Price



From the above graph, we see a very high correlation between the property price and loan amount requested. It is safe to assume the more expensive a property is, the higher the amount of loan is required to purchase it

Loan Amount Requested vs Loan Sanction Amount



The loan amount requested increases with the loan sanctioned amount.

Higher the loan amount you ask for, the more likely you will be able to receive a large sum from the bank

Conclusion

We have preprocessed the data, dropped/filled missing values and removed outliers using various standard statistical data preprocessing techniques. With our cleaned data, we have performed an exploratory data analysis to analyze underlying patterns and correlations. Finally, we found out that the dependent variable Y i.e, the loan sanctioned amount is determined mainly by the loan amount requested, credit score, property price, current loan expenses, loan amount request and income. The least important features are no. of defaults, age, property type and gender.

Based on our complete analysis, we will drop less important features which are less correlated with the target variable and standardize the data before fitting our machine learning model.

PHASE 2:

Data Preprocessing for model implementations:

- **Objective:** The goal of preprocessing is to prepare the dataset for machine learning by addressing irrelevant features, skewness, scaling issues, and ensuring numerical compatibility for the model.
- **Dropping Irrelevant Features:** First, we have dropped the features which are irrelevant. The dropping of the feature is based on the correlation between different features. From our EDA analysis, it was clear that these columns are not highly correlated with the target variable: 'Customer ID', 'Name', 'Type of Employment', 'Property ID', hence they are dropped.
- **Handling Skewness:** We calculated the skewness of the numerical columns. Highly skewed features (Property Age and Income (USD)) were transformed using logarithmic scaling ($\log(1 + x)$), and moderately skewed features (No. of Defaults) were transformed using the square root function.
- Skewed distributions can lead to models being biased toward certain ranges of data. So handling this skewness is necessary.
- **Encoding Categorical Variables:** Categorical columns (['Gender', 'Profession', 'Location', etc.]) were converted into numerical representations using LabelEncoder. This is done because Machine learning models require numerical data as input. Label encoding transforms categorical data into integers, making it suitable for models while preserving unique category distinctions.
- **Splitting Features and Target Variables:** The dataset was split into features (**X**) and target variable (**y**). The target variable was reshaped to meet the requirements of subsequent scaling operations. Separating features and target ensures clarity and compatibility with supervised machine learning tasks.
- The dataset was divided into training (80%) and testing (20%) sets. Using 80% for training allows the model to learn effectively while leaving enough data for a good evaluation.
- **Scaling Features and Target Variables:** StandardScaler was applied to both features (**X**) and the target variable (**y**). Standardization ensures all features have zero mean and unit variance, which is crucial for gradient-based algorithms and distance-based models. Scaling the target variable makes its range consistent with the feature scales, preventing one from dominating the model's optimization process.

Model Implementations: We have implemented the following models:

1. Gradient Boosting

2. Random Forest
3. Linear Regression
4. Neural Network
5. Principal Component Analysis (PCA) Regression

1. Gradient Boosting:

Why Gradient Boosting?

- Predictive Power: Captures complex relationships and patterns, making it ideal for tabular data like loan sanction predictions.
- Flexibility: Effectively handles both numerical and categorical features with minimal preprocessing.
- Robustness: Hyperparameter tuning helps balance bias and variance, reducing the risk of overfitting.

Implementation:

- Data Preprocessing: The dataset was cleaned, categorical features were encoded, and data was split into training/testing and then scaled using the StandardScaler.
- Model Training: The GradientBoostingRegressor from Scikit-Learn is utilized
- Hyperparameter Tuning: Grid Search is applied to optimize parameters for better performance.
- Evaluation: Model performance was assessed using RMSE (8864.41) and R^2 (0.9515). SHAP was used to analyze feature importance.

Key Parameters:

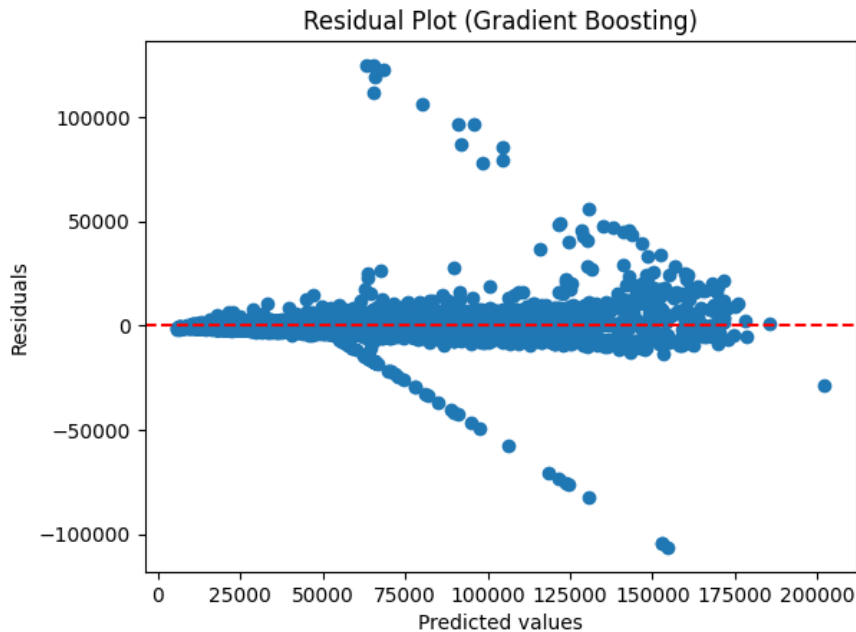
- learning_rate: 0.05 (controls the contribution of each tree).
- n_estimators: 200 (number of trees).
- max_depth: 5 (limits the complexity of each tree).
- subsample: 1.0 (uses all samples).
- max_features: None (uses all features).
- Gradient Boosting was chosen for its strong predictive performance and its ability to manage the complexity of the data effectively.
- These parameters strike a balance between underfitting and overfitting, ensuring the model generalizes well to unseen data.

Performance Metrics:

- The RMSE value of 8864.41 USD is relatively low considering the range of loan sanction amounts in the dataset. This suggests that the Gradient Boosting model is performing well, providing predictions that are reasonably close to actual values on average.
- R^2 Score: 0.9515
- The R^2 score suggests that approximately 95.15% of the variance in the loan sanction amount is explained by the model. This demonstrates a strong fit of the model to the data.

Visualization:

1. Residual Plot:



Key Observations:

- Strengths:

- Residuals are reasonably distributed around the red dashed line, indicating good overall predictive performance.
- Higher predicted values show fewer extreme deviations, demonstrating the model captures general trends well.

- Areas for Improvement:

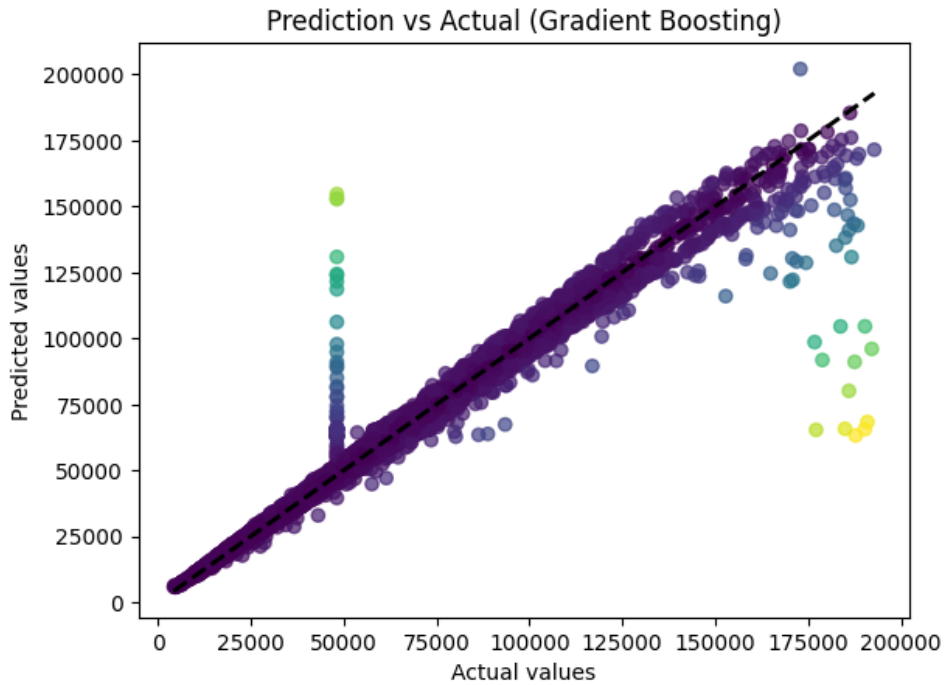
- **Non-Uniform Spread:** A funnel shape for lower predicted values suggests **underprediction** for high actual values.
- **Outliers:** Some extreme negative residuals (bottom-left) indicate the model struggles with specific cases, particularly underpredicting certain loans.
- **Systematic Pattern:** Diagonal clustering in negative residuals hints that some feature interactions are not fully captured.

- Implications:

- The model performs well overall, but issues like **heteroscedasticity** (increasing variance of residuals with higher predicted values) and occasional bias warrant further attention.
- Improvements could include feature engineering, addressing outliers, or exploring advanced models to better handle complex patterns.

This plot reflects a generally strong model with clear opportunities for refinement.

2. Prediction vs Actual Plot:



- It can be seen that the most of the loan amount is being predicted accurately and the trend is being captured.

3. SHAP Summary Plot



1. Feature Importance

- Features are ranked by their impact on the model's prediction (most important at the top).
- "Loan Amount Request (USD)" is the most impactful feature.

2. SHAP Values (X-axis)

- Negative SHAP values (left) reduce predictions, while positive values (right) increase them.

3. Insights

- **"Loan Amount Request (USD)"**: High values (red) increase predictions.
- **"Credit Score"**: High scores (red) decrease predictions (e.g., reduced risk).
- Features like "Gender" or "Property Type" have minimal impact (values close to zero).

4. Purpose

- Highlights feature importance and how feature values influence predictions.

2. Random Forest:

Why Random Forest?

- Primarily, our loan dataset is a real world dataset which doesn't exhibit a strictly linear relationship between the deciding factors/features and the loan amount sanctioned. Random forest has the ability to capture intricate feature relations and model such complexly related datasets.
- Secondly, a random forest model can give additional information such as which factors most significantly impact the target variable, which can add meaning to the decision making process for the bank on the loan amount sanctioned in case there is a secondary manual decision process.
- They are more robust to extreme values which can't be removed as part of outliers. Such values don't make the model skewed. Our data which has many such instances, therefore benefit from a random forest regressor.
- Random Forest is an ensemble model, meaning it builds multiple decision trees each of which can make a prediction, and the final result is an average of these predictions, which means there is resistance against overfitting.
- Random Forest has multiple hyperparameters which can be hypertuned to render an optimal combination of hyperparameters that yield best results. Thus we're able to get a good performing model to predict loan amounts sanctioned.

Implementation and Key Parameters:

- Using GridSearch CV library, we implemented hyperparameter tuning to determine the most ideal hyperparameter values for training the model.
- We generate a best_model which is trained using these values
- We implemented a 5-fold cross validation using a subset of training data by creating training and validation data in the ratio 4:1, to prevent overfitting. This was done for 5 fractions of training data to create more randomness. For each fraction of training data, 5-fold cross validation was implemented on it and the mean and the standard deviation of Mean Squared Error/MSE values across all folds was calculated. This mean and standard deviation of MSE

was plotted for all fractions of the data. Using data fractions will help us determine if adding more data can improve model performance, and cross validation reduces overfitting.

- The final MSE for this validation was **0.06** and standard deviation was **0.01**, which is a highly negligible value
- MSE was calculated using the test_data. The result was **0.5**, which is lesser than corresponding metrics on validation data. It indicates that the model is not overfitting or underfitting.
- Because we initially scaled down the target variable, we reverse the transformation on y_test and also on y_pred, and we calculate the Root Mean Square Error on this, as the original values of target are extremely large(5-6 digits), and thus the associated MSE and MAE values are large
- RMSE in Original Scale: **8809.82**
- Mean Absolute Error: **3803.48**
- R2 score: **0.95**

Performance Metrics

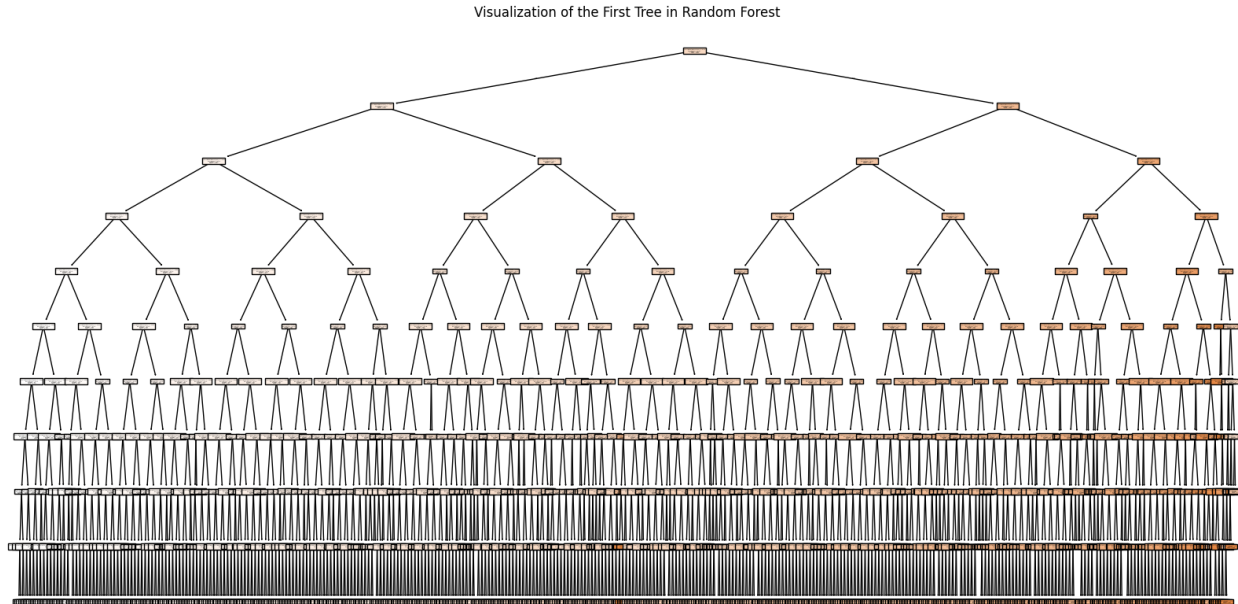
RMSE in Original Scale: **8809.82**

R2 score: **0.95**

- The RMSE value of 9141.60 USD is relatively low considering the range of loan sanction amounts in the dataset. This suggests that the Random Forest model is performing well, providing predictions that are reasonably close to actual values on average.
- The R² score suggests that approximately 95 percent of the variance in the loan sanction amount is explained by the model. This demonstrates a strong fit of the model to the data.

Visualisations:

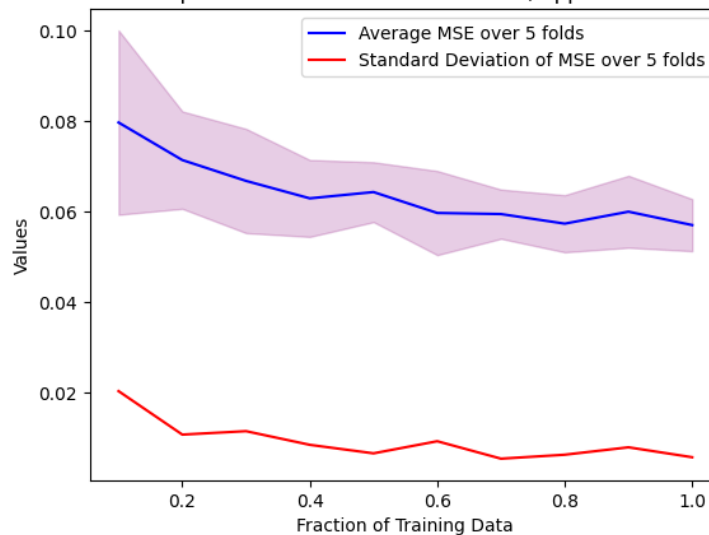
1. Sample tree generated:



- We have a binary tree, with depth = 10,

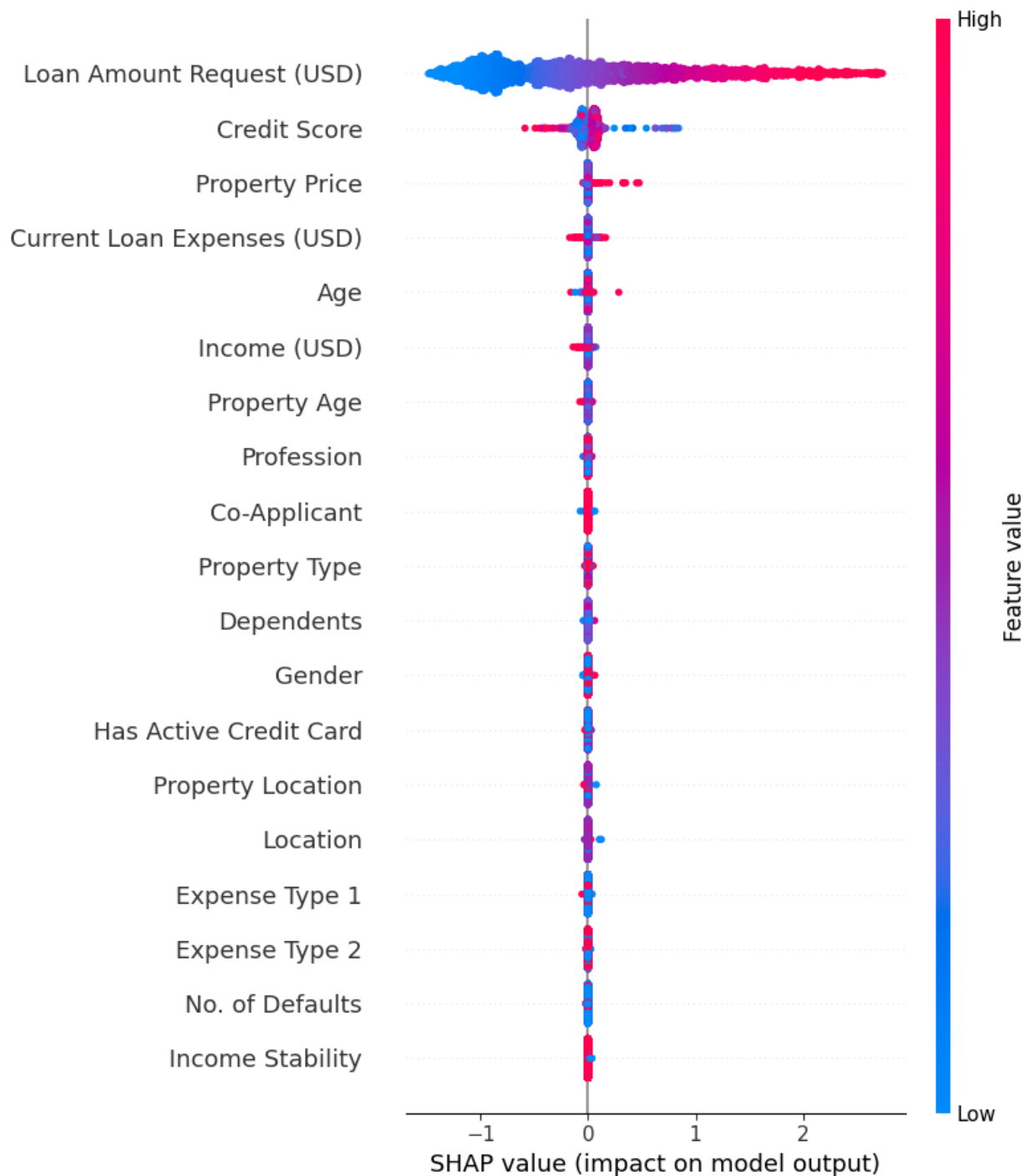
2. Mean and Standard Deviation of MSE over each iteration of 5-fold cross validation, calculated for different fractions of training data size.

Mean and Standard Deviation of MSE per fold of 5-fold cross validation, applied on different fractions of training data



- As the size of training data increases, the average MSE over all iteration 5 iterations for 5-fold cross validation decreases. This indicates that increasing the amount of training data is helping the model learn better and reduce errors.
- Similarly, standard deviation of MSE across all iteration 5 iterations for 5-fold cross validation also decreases suggesting that the MSE is consistent for each of the 5 epochs for maximum amount of training data.
- Thus this plot aids in detecting data sufficiency and performance of model on validation data.

3. SHAP summary Plot

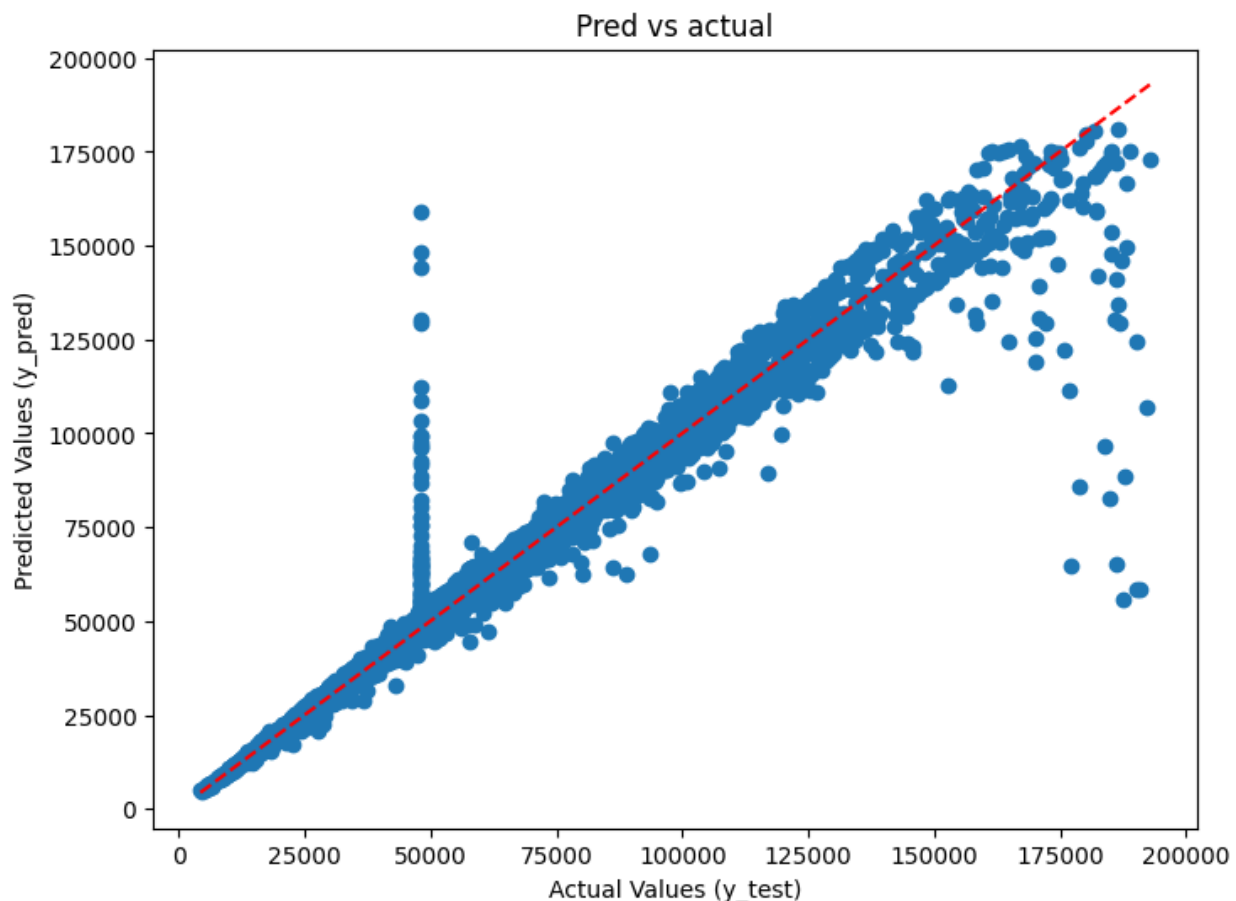


- This SHAP plot provides a more detailed analysis of impact of each feature on the prediction outcome of the model. The x-axis represents the SHAP values, indicating how much each feature pushes the model's prediction positively (right) or negatively (left). The color code

indicates what extremity (low or high) of each feature's values impact the model positively or negatively.

- Some notable inferences are:
- High loan request amounts (red) have a strong positive impact on the model's prediction, and implies that in most cases the requested amount is delivered.
- High credit score pushes the model to predict higher values of loan amount sanctioned, possibly reducing risk or increasing approval chances.
- High property prices pushes the model to predict higher loan amount sanctioned.
- High current loan expenses tends to negatively affect the outcome, whereas low current expenses impacts it positively.
- All other features are not contributing much to the outcome.

4. Actual vs Predicted Values



- We can see that the majority of data points are being predicted correctly. Some values which are actually lesser in magnitude are wrongly predicted as high values. This

might possibly be because of insufficient representation of these cases in the training dataset. This can be handled by collecting more data, in the real world scenario.

- The incorrect values contribute heavily to the error because of the magnitude of the target variable.

3. Linear Regression:

Why Linear Regression:

- Linear Regression is a suitable model for loan prediction analysis due to its simplicity, interpretability, and efficiency. It provides meaningful insights into relationships between features and outcomes, ensuring that the results are understandable. From our EDA analysis we analyzed that many features have linear relationship with the target variable, hence choosing linear regression makes sense.

Implementation:

- A LinearRegression model is instantiated and trained on the scaled training data using sklearn's LinearRegression model.
- The model was trained on scaled training data.
- Mean Squared Error (MSE) and Root Mean Squared Error (RMSE) were computed for scaled and original data.
- R^2 score was calculated to measure the proportion of variance explained by the model.
- Residual plots and scatter plots were generated to visualize the fit.
- 5-fold cross-validation was applied to evaluate model performance across different fractions of training data. The average MSE and its standard deviation were plotted.

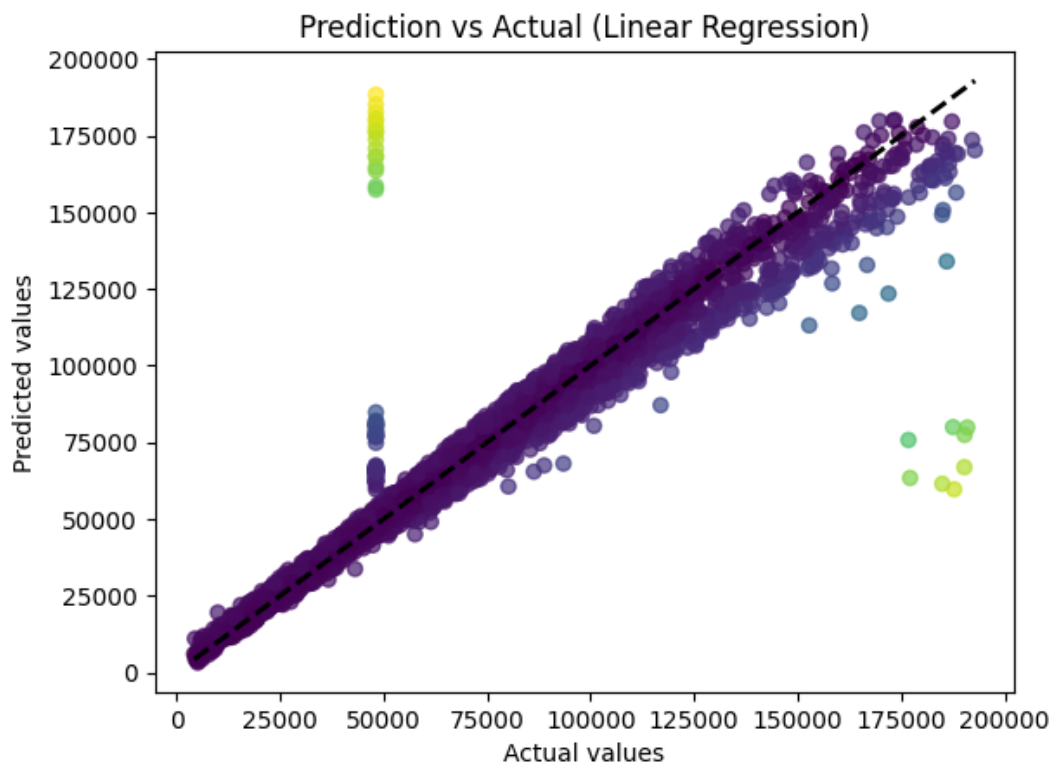
Key Parameters and Performance Metrics

- Linear Regression in Scikit-learn has no hyperparameters to tune, making it straightforward to use. The simplicity allows focusing on data preprocessing and evaluation.
- MSE value: 0.08. This indicates the model performs well on scaled data.
- RMSE value: 11077.69. This is a useful interpretation of how much error, on average, is expected in predictions. The error in original scale is reasonable given the target variable's range.
- R^2 score: 0.9243. This shows that ~92.4% of the variance in the target is explained by the model, indicating strong predictive performance.
- Average MSE across 5 folds: 0.08. This indicates consistent performance across different subsets of training data.
- Standard Deviation of MSE across 5 folds: 0.01. A low standard deviation highlights the model's stability and reliability.

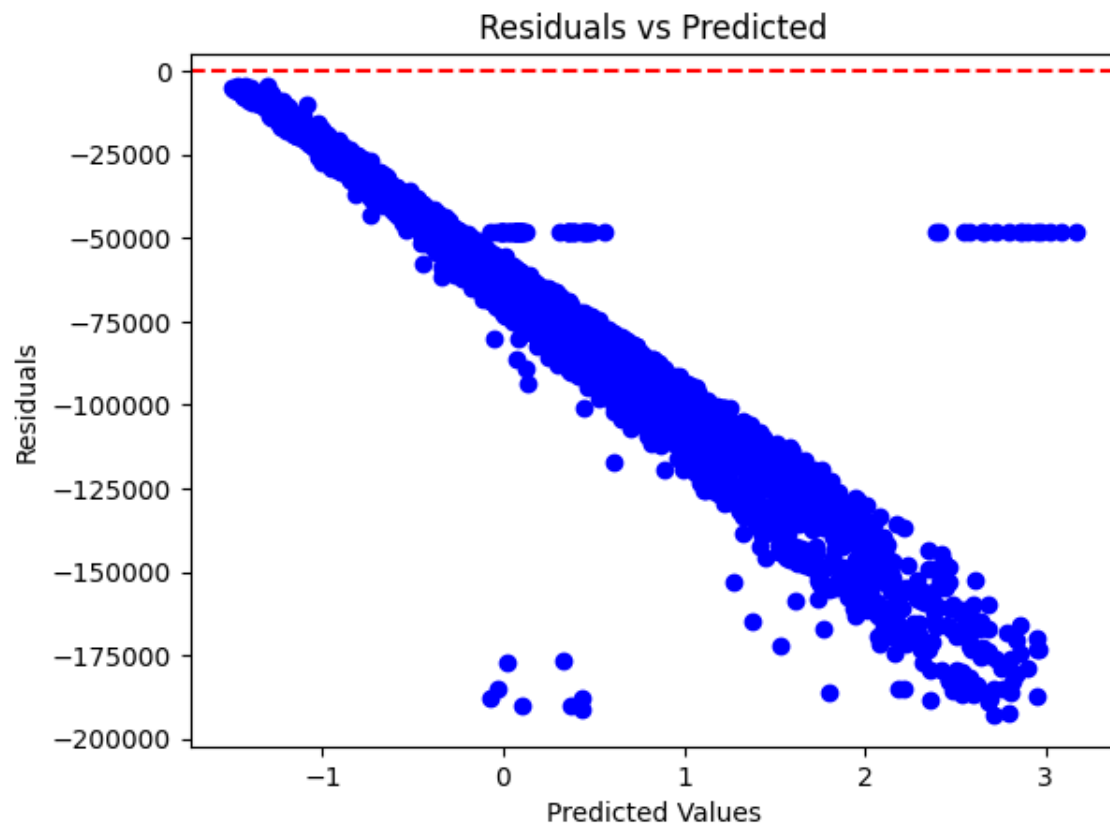
Visualization:

1. Prediction vs Actual Plot:

- It can be seen that the most of the loan amount is being predicted accurately and the trend is being captured using this model.



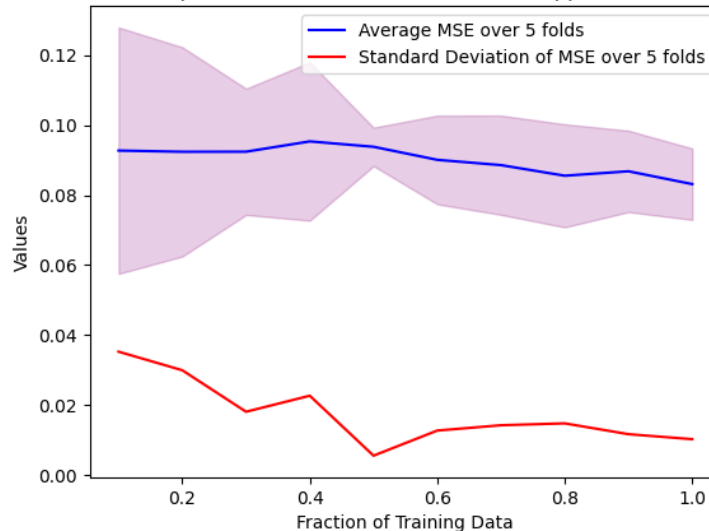
2. Residuals vs. Predicted:



- **Non-linear Trend:** The residuals show a distinct downward slope rather than being randomly distributed around zero. This indicates the model may be misfitting and is potentially failing to capture the underlying pattern of the data. This shows that the real world data is not linear and this model fails to capture the complexities of these features. Hence is poorly fitted.
- The spread of residuals varies across the predicted values. This suggests that the variance of errors is not constant, which could violate assumptions of linear regression or similar models.

3. Mean and Standard Deviation of MSE per Fold

Mean and Standard Deviation of MSE per fold of 5-fold cross validation, applied on different fractions of training data



- **Convergence with More Data:** The average MSE (blue line) stabilizes as the fraction of training data increases, indicating that the model's performance improves with more data and becomes more consistent.
- **Low Variability:** The standard deviation (red line) decreases as the fraction of training data increases, showing that the model's predictions are more reliable and less sensitive to variations in training data.
- **Early Variability:** At smaller fractions of training data, the model exhibits higher variance and MSE, indicating that it struggles to generalize well with limited data.

4. Neural Network:

Why Neural Network:

- We have chosen a neural network for this task, since it can model complex, nonlinear relationships between input features and the target variable. NNs are good at detecting complicated patterns in big data, especially when the use of linear models would not be appropriate.
- Large amounts of neurons and several layers allow the model to extract higher-order features step by step from the raw data to make its predictions. Besides that,

we applied the regularization technique, including L2 regularization and dropout to avoid overfitting so that the model generalizes well on unseen data.

Implementation:

- Used the Keras Sequential API to define the architecture of the neural network.
- Added several dense layers with ReLU activation functions, which are suitable for capturing non-linearities in the data.
- Incorporated regularization techniques:
 - o L2 regularization: Applied to certain layers to penalize large weights and avoid overfitting.
 - o Dropout: Added to randomly set 30% of the input units to 0 during training to prevent the model from relying too much on specific features.
 - o Batch Normalization: Used to normalize activations and improve training speed and stability.
- The model was compiled using the Adam optimizer, a popular choice for deep learning models, with the mean squared error (MSE) loss function. Mean absolute error (MAE) was used as an additional metric.
- Trained the model for 50 epochs with a batch size of 16, employing early stopping with validation data to prevent overfitting and ensure the best model performance.

Key Parameters:

- Number of Layers: The network has 7 dense layers with varying numbers of units (64, 128, 64, 64, 32, 32, and 16) to capture different levels of abstraction in the data.
- Activation Function: ReLU was used in all hidden layers, as it's effective in preventing vanishing gradients and speeding up training.
- Regularization:
 - L2 regularization with varying strength (0.1 to 0.5) was applied to specific layers to prevent overfitting.
 - Dropout (30%) was used after certain layers to reduce the risk of overfitting and enhance generalization.
- Optimizer: Adam optimizer, known for its adaptive learning rate and efficient handling of sparse gradients.
- Loss Function: Mean squared error (MSE) was chosen for regression tasks, as it penalizes larger errors more significantly.
- Metrics: Mean absolute error (MAE) was used to provide an alternative evaluation of the model's performance.

4. Results after visualization

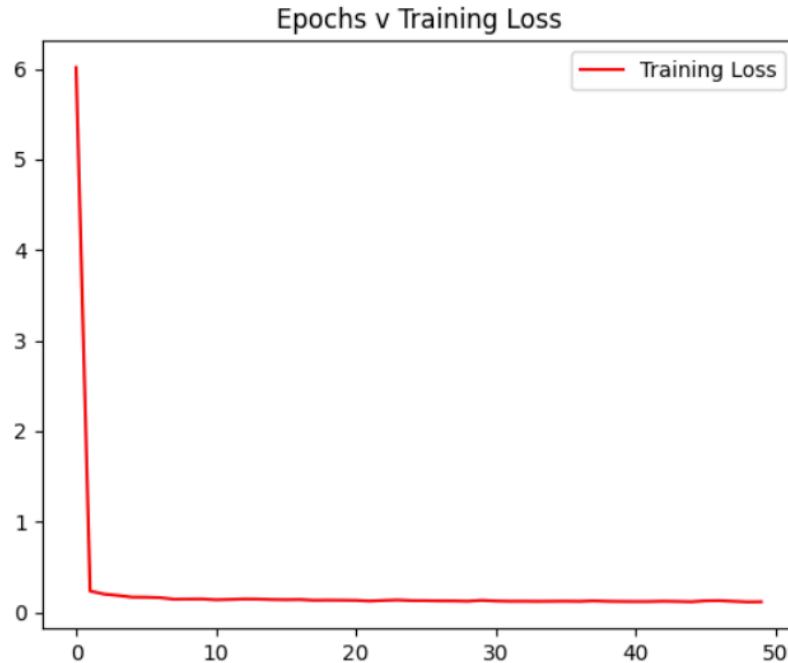
- R^2 : The coefficient of determination (R^2) value of 0.853 indicates that the model explains approximately 85.3% of the variance in the test data, which is a strong result for a regression task.
- MSE and RMSE: The value of 0.151 MSE with scaled data reflects the average squared difference between predicted and actual values, with lower values indicating better performance and value of 15441.22 RMSE for unscaled data. The high value in RMSE is due to the magnitude of the target variable being high
- MAE: 0.2827 (Scaled Data)

The mean absolute error shows the average magnitude of errors in the model's predictions, with a relatively small value indicating good accuracy.

Visualization:

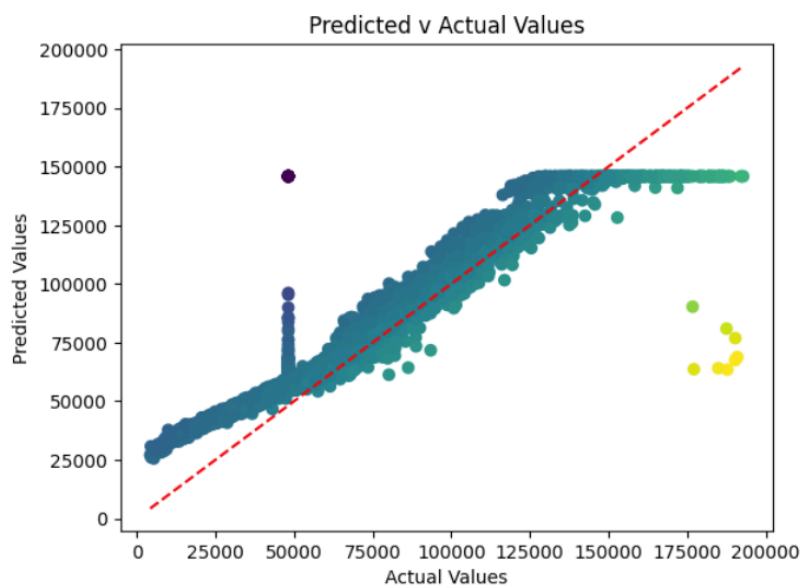
1. Training Loss vs Epoch:

- The training loss curve shows a steady decrease, indicating that the model is learning effectively. Early convergence suggests that the chosen architecture and hyperparameters are appropriate for the task.



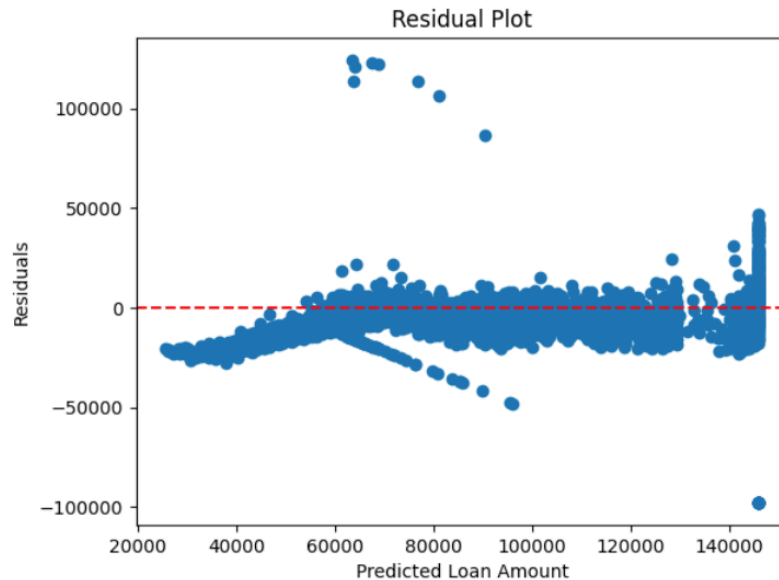
2. Predicted vs Actual Values:

- The scatter plot shows that the predictions are closely aligned with the actual values, confirming that the model generalizes well to the test data.



3. Residual Plot:

- The residuals plot (predicted vs residuals) shows no significant patterns, implying that the model's errors are random and not indicative of any systematic biases.



5. PCA Regression:

Why PCA Regression:

- Our dataset, the loan dataset has multiple features which increases the dimensionality of the dataset. Most of these features have multicollinearity and are correlated. It also has some random noisy features like Expense Type 1 and Expense Type 2. PCA transforms the data into a smaller set of **principal components** which will therefore help capture most of the variance, simplifying the model and reducing computational complexity.

Implementation:

- PCA is initialized to reduce the features to 10 principal components
- Dataset is split into test and train and then scaled using standard scaler.
- PCA is applied to the scaled training features:
- Fit and transform the training data: `pca.fit_transform`.
- Transform the test data using the same PCA model: `pca.transform`.
- A linear regression model is trained on the reduced-dimensional training data and scaled target values
- Test data is used to predict loan sanction amount values

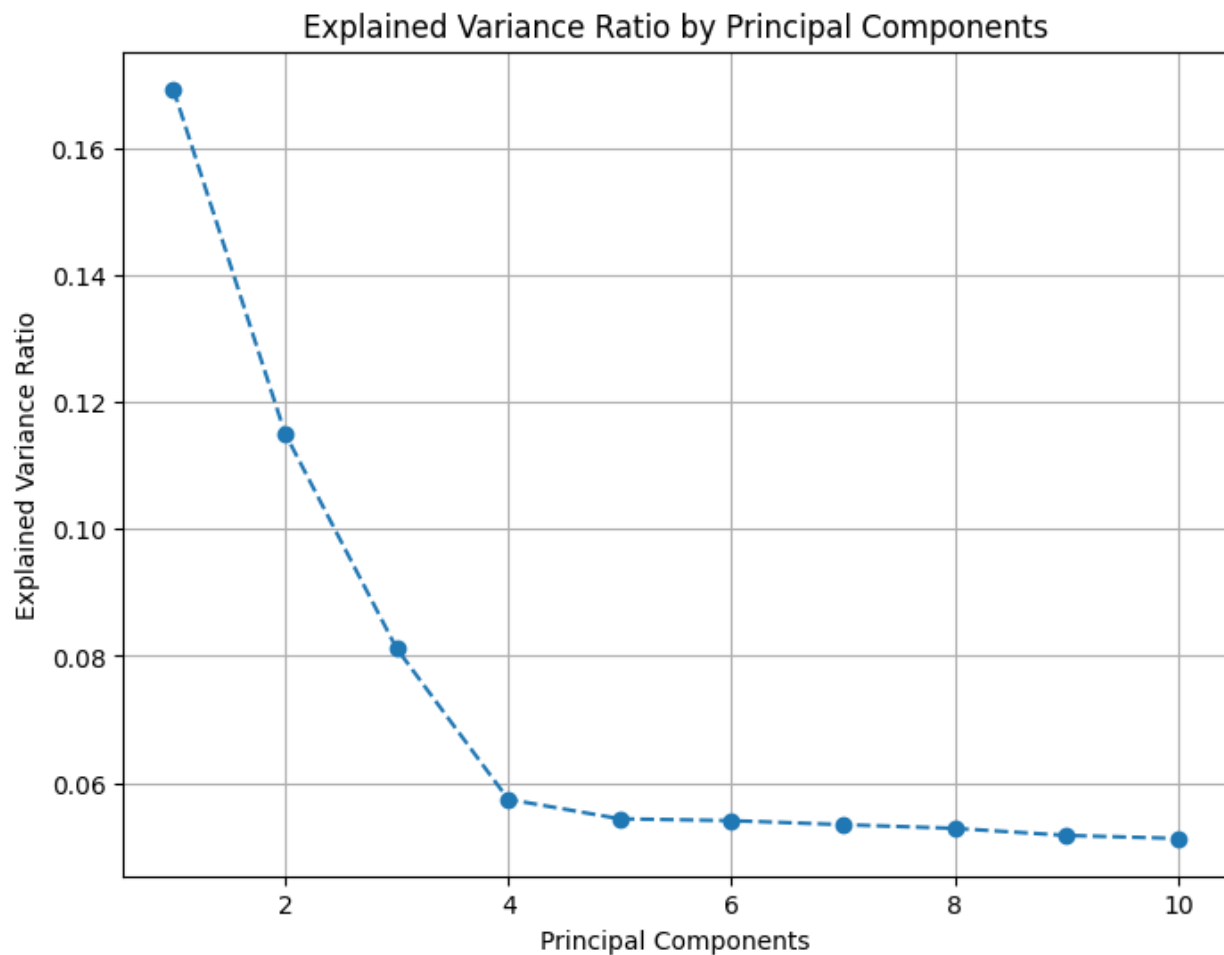
Performance Metrics:

- Mean Squared Error: 214260567.15889716
- R^2 Score: 0.8678853983014863

- The high R^2 score indicates the model captures most of the variation in the target variable.
- The model likely benefits from the dimensionality reduction via PCA, as it avoids overfitting to irrelevant features.
- The high MSE is likely because the magnitude of the target variables are huge and compared to other models, this PCA linear regression model is not able to make as many accurate predictions as other models.

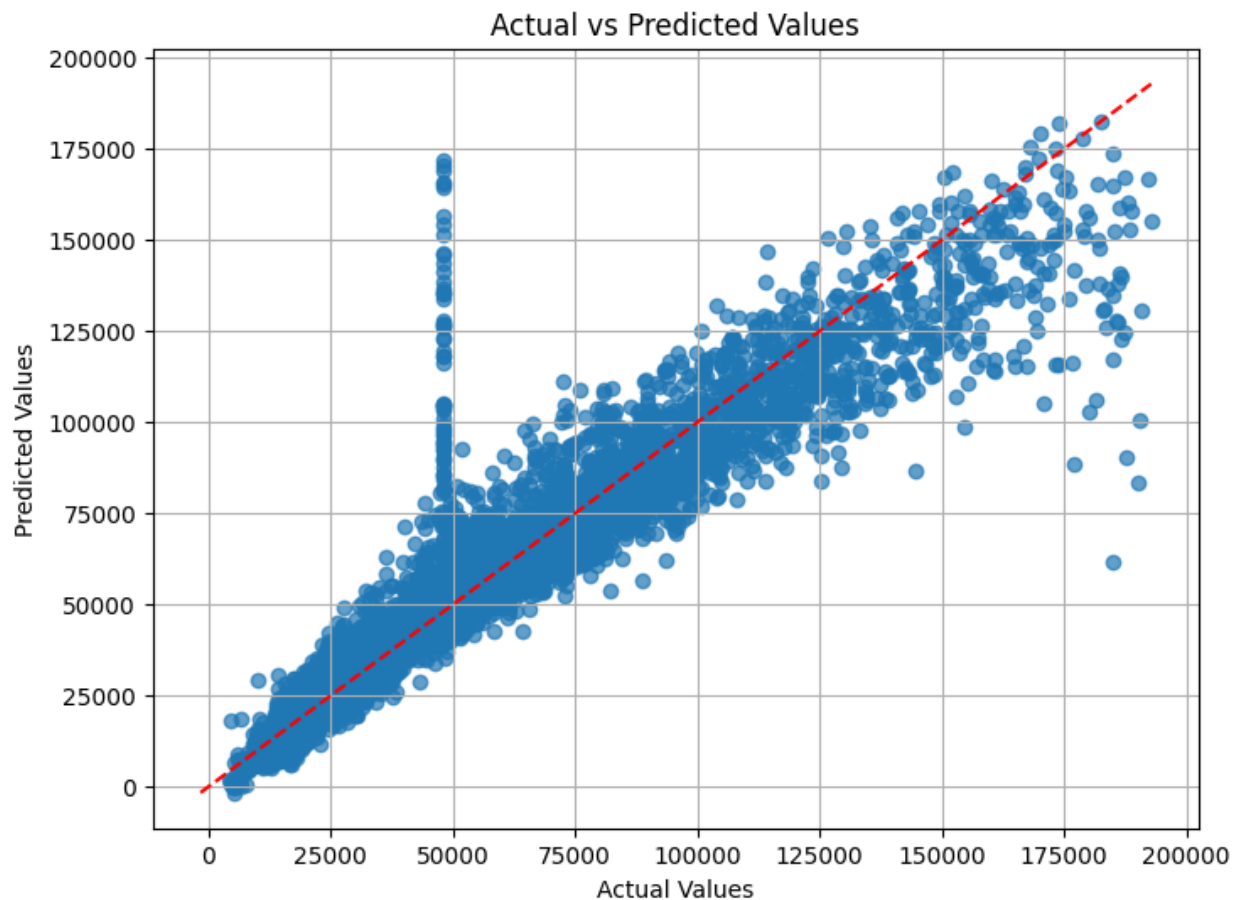
Visualisation

1. Explained Variance Ratio by PCA



- **Steep Decline in Variance:** The first few principal components explain the majority of the variance in the dataset. Specifically, the first component explains about 16%, and the second adds another ~12%.
- After the 4th component, the explained variance plateaus, with subsequent components contributing only marginally.
- The "elbow" in the curve which is around **3–4 components** suggests the optimal number of components for capturing most of the variance without including unnecessary dimensions.
- This tells us that instead of keeping all 10 components, **3–4 components** might suffice to retain the most meaningful information.

2. Actual vs Predicted



- The predicted values are mostly close to the actual values, with few exceptions with major difference, which is likely contributing to the high Mean Squared Error value, despite a good R^2 score.
- Residuals vs Predicted Values
- The residuals (differences between actual and predicted values) are not uniformly scattered around the red line (zero residual). A noticeable **funnel shape** is visible, with the variance of residuals increasing as predicted values increase.
- The spread of residuals is not constant across all predicted values, indicating **heteroscedasticity**. This violates a key assumption of linear regression, which assumes homoscedasticity.
- For lower predicted values, the residuals are mostly negative ie, model tends to **overpredict**. For higher predicted values, residuals show both large positive and negative errors ie, model struggles with extreme values.
- Hence despite PCA because of the non linear nature of data probably, the PCA based linear regression model is not highly effective.

Explanation of Metrics:

- **MSE (Mean Squared Error):** Measures the average squared difference between predicted and actual values. Lower is better.
- **RMSE (Root Mean Squared Error):** Square root of MSE, representing error in the same units as the target variable. Lower is better.
- **R² Score (Coefficient of Determination):** Indicates the proportion of variance in the target variable explained by the model. Higher is better, with a maximum value of 1.

Model	MSE	RMSE	R ² Score
Gradient Boosting	0.25	9157.37	0.948
Random Forest	0.05	3854.27	0.95
Linear Regression	0.08	11069.533	0.924
Neural Network	0.1511	15441.22	0.85
PCA Regression	0.135	14637.64	0.867

Analysis of Each Model:

Gradient Boosting:

- **MSE:** 0.25 (High error compared to Random Forest but better than Neural Network and PCA Regression).
- **RMSE:** 9157.37 (Moderate error in real-world units).
- **R² Score:** 0.948 (Second-highest, indicating excellent explanatory power).
- **Analysis:** Gradient Boosting performs well but is slightly outperformed by Random Forest in error metrics. It strikes a good balance between simplicity and accuracy.

Random Forest:

- **MSE:** 0.05 (Lowest among all models, indicating the least average error).
- **RMSE:** 3854.27 (Smallest real-world error).
- **R² Score:** 0.95 (Highest, meaning it explains the most variance in the target variable).
- **Analysis:** Random Forest is the best performer in all metrics, with a perfect balance of minimal error and maximal explanatory power.

Linear Regression:

- **MSE:** 0.08 (Better than Gradient Boosting but worse than Random Forest).
- **RMSE:** 11,069.53 (High error compared to most other models).
- **R² Score:** 0.924 (Good but the lowest among non-PCA-based models).
- **Analysis:** Linear Regression has decent performance but lags behind both Random Forest and Gradient Boosting.

Neural Network:

- **MSE:** 0.1511 (Better than PCA Regression but worse than Linear Regression).
- **RMSE:** 15,441.22 (Highest real-world error, making it the least reliable in predictions).
- **R² Score:** 0.85 (Lowest explanatory power among all models).
- **Analysis:** Neural Network performs poorly in this setup. It might require hyperparameter tuning or more data to improve performance.

PCA Regression:

- **MSE:** 0.135 (Better than Neural Network but worse than all others).
- **RMSE:** 14,637.64 (Second highest, indicating significant prediction error).
- **R² Score:** 0.867 (Low explanatory power, only slightly better than Neural Network).
- **Analysis:** PCA Regression doesn't perform well, likely because dimensionality reduction using PCA sacrifices important variance.

Best Model:

The Random Forest model is the best among the five, as it:

- Has the lowest MSE (0.05), indicating minimal average error.
- Achieves the lowest RMSE (3854.27), suggesting the most accurate predictions in real-world units.
- Scores the highest R² (0.95), demonstrating its ability to explain 95% of the variance in the target variable.

Conclusion:

- Our model provides highly accurate loan amount predictions, enabling users to deliver real-time feedback to customers.
- Improvements -
 - Each model performs optimally when predicting large loan amounts.
 - Additional data enhances the model's performance.
 - We can incorporate more variations and features to further improve accuracy and utility.
 - Adding more variations and features, such as demographic, economic, or behavioral attributes, can capture nuanced patterns, leading to more robust and reliable predictions.
 - Continuous feature engineering, including dynamic data sources and contextual information, will help adapt the model to evolving trends in loan applications and customer behavior.
- **This prediction system which gives nearly accurate results, can be used by both the customers and the bank officials to make a preliminary estimate of the loan amount a customer is eligible for given their personal and financial history.**